Machine Learning Study Flow(Regression)

1. ML Foundations

What is ML?

→ Classical programming vs ML
→ Supervised vs Unsupervised
→ Features and target
→ Training and testing

↓

2. Regression

Simple Linear Regression

→ Equation
→ Prediction
→ Error/residual
→ Loss function
→ Cost function (MSE)

↓

3. Multiple Linear Regression

More than one feature

→ Weights + bias
→ Model equation
→ Coefficient interpretation
→ Train/test split
→ Predictions
→ Regression evaluation metrics

↓

4. How does the model learn?

Gradient Descent

→ Cost function
→ Gradient
→ Learning rate
→ Weight update

→ Iterations
→ Convergence

↓

5. Feature Scaling

Why do features need comparable scales?

→ Feature magnitude
→ Standardization
→ Normalization
→ StandardScaler
→ Effect on Gradient Descent

↓

6. Polynomial Regression

What if a straight line is too simple?

x → x,x^2,x^3

→ Polynomial features
→ Model complexity
→ Underfitting
→ Overfitting

↓

7. Overfitting → Regularization

→ Regularization idea
→ Penalty term
→ Ridge / L2
→ Lasso / L1
→ Alpha / λ
→ Coefficient shrinkage
→ Lasso → coefficients can become zero